DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 3

Web developer

Profession 3 of 10 in Development and IT.

What the profession means

A web developer builds websites: the pages, layout and interactivity that visitors use. A web developer usually works with HTML, CSS, JavaScript and sometimes a content system or framework.

What professionals do every day

Turns designs into working pages, makes them responsive and accessible, connects forms and analytics, tests across browsers and devices, and publishes the site.

Why businesses hire them, and what problems they solve

Customers judge a business by its website in seconds. A developer makes the site fast, readable on phones, easy to contact the business through, and found by search engines.

- Designs that look good but break on a phone.
- Pages that are slow or hard to use for people with disabilities.
- Forms that lose enquiries.

Skills you need

- HTML semantics, CSS layout (flexbox and grid), responsive design.
- JavaScript for interaction and form checks.
- Accessibility basics (labels, contrast, keyboard use).
- Performance basics (image size, caching).
- Publishing and hosting.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Visual Studio Code	Code editor	Free	Free	code.visualstudio.com
Chrome DevTools	Inspect, debug and test layouts	Free	Free	developer.chrome.com/docs/devtools
Netlify	Publish static sites	Free tier	Paid plans	www.netlify.com
GitHub Pages	Free hosting for static sites from a repository	Free	Free	pages.github.com
W3C validator	Checks HTML errors	Free	Free	validator.w3.org

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Static pages with HTML and CSS; responsive layouts; a simple form with checks.
Intermediate	JavaScript interactions, accessible components, forms that save to a service.
Advanced	Performance tuning, a component library, analytics and an automated test.

Projects by level

- **Beginner:** A four-page site for a local service business.
- **Intermediate:** A booking page with a calendar and a working contact form.
- **Advanced:** A content-heavy site with search, filters and measured performance.

Client-ready portfolio project

A responsive, accessible, search-friendly site for a real or clearly fictional local business, with a performance report and a handover page.

Worked example

Worked example. Turning a design card into HTML: a card is a `<article>` with a heading, a paragraph and a link. CSS gives it padding and a border; the grid places the cards.

Assignment with full solution

ASSIGNMENT

A responsive business landing page

Business problem: A plumbing company needs a single landing page with services, prices and a call button that works on a phone.

Steps

1. Build a header with the company name, a 'Call now' link using `tel:`, and a menu of three services.
2. Add a services section with three cards that stack on a phone and sit in a row on a laptop.
3. Add a contact form with name, phone and service fields; label every input.
4. Test at 390 and 1280 pixels and fix any horizontal scrolling.

Expected output: One HTML page, one CSS file, accessible labels, and no horizontal scrolling at 390 pixels.

SOLUTION (ANSWER KEY)

Solution outline: the call link uses `<a href="tel:+919800000000">`, which opens the phone dialler on a mobile device. The cards use CSS grid with `grid-template-columns: 1fr` by default and `repeat(3, 1fr)` inside a media query for widths of 720 pixels and above. Each input has a visible <label>, which screen readers and autofill rely on.

CSS

```
.cards { display: grid; gap: 16px; grid-template-columns: 1fr; }
@media (min-width: 720px) { .cards { grid-template-columns: repeat(3, 1fr); } }
a.call { display: inline-block; padding: 12px 18px; background: #6c4cf5; color: #fff; border-radius: 10px; text-decoration: none; }
```

Test every page at 390 and 1280 pixels, check that all images have alt text, and validate the HTML with the free W3C validator (validator.w3.org).

Common mistakes

- Using `<div>` for everything instead of meaningful elements.
- Forgetting to test on a phone until the end.
- Unoptimised images that slow the page.
- Forms without labels or error messages.

Freelancing opportunities

- Small business landing pages and brochure sites.
- Fixing and redesigning existing websites.
- Freelance marketplace briefs for 'landing page', 'HTML/CSS', 'responsive website'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- web.dev Learn (Google)
- MDN: Learn web development
- W3C Web Accessibility tutorials

Part of **Master Freelancing: Development and IT**. Back to the chapter list on skillnavora.in · Open the full textbook. Official links in this chapter were not checked live; confirm before relying on them.